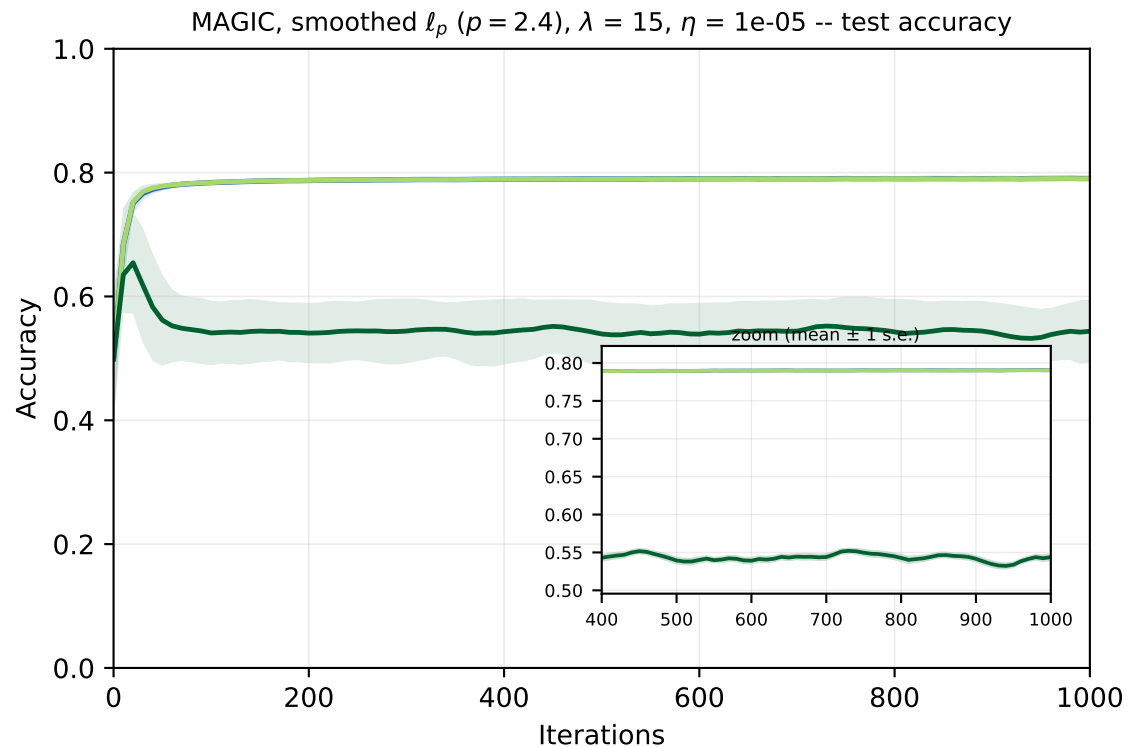
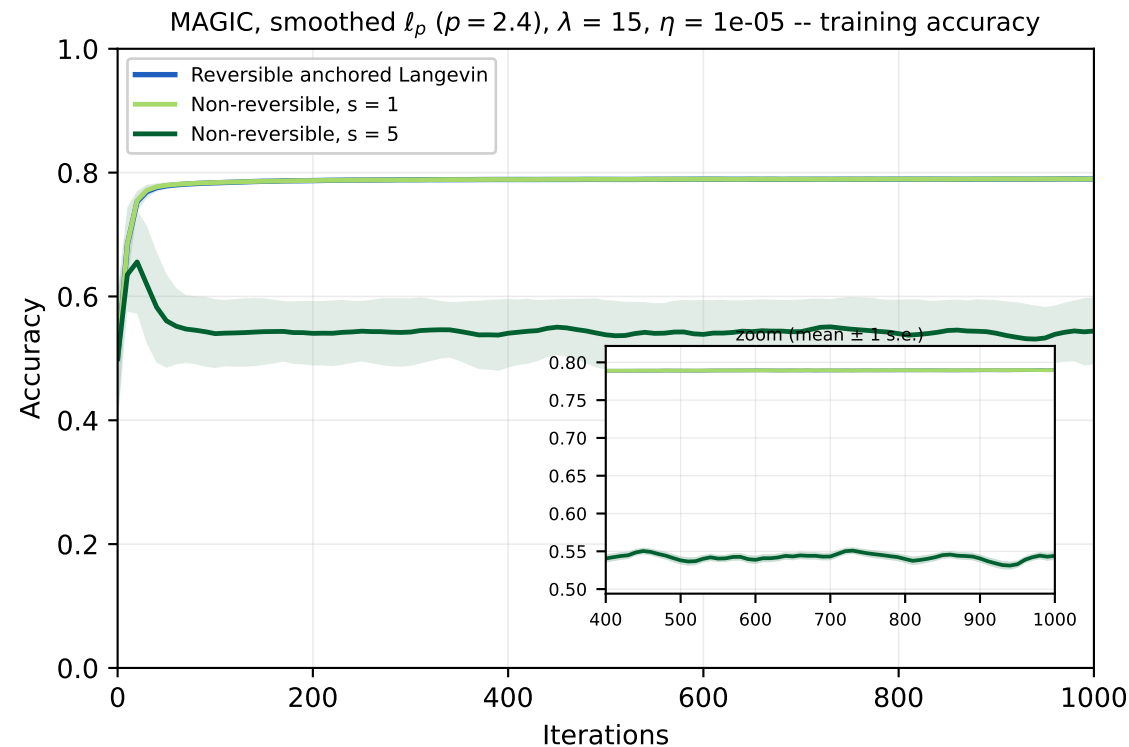


magic_lp: $\lambda = 15$, $\eta = 1e-05$



MAGIC -- smoothed ℓ_p $g(w) = \sum_i (w_i^2 + \text{eps}^2)^{p/2} \leq \text{Lambda}$, $p = 2.4$, $\text{eps} = 0.2$, $\text{Lambda} = 4$. $d = 10$ (intercept $w_0 + 9$ features). $U = \sum \text{softplus}(-y x \cdot w + w_0^2 / (2 \sigma^2) + \lambda \sum_{j=1}^d |w_j|$; U_0 replaces $|w_j|$ by $\sqrt{w_j^2 + \delta^2}$; $a = \exp(U - U_0)$. EXACT gradient. $\sigma = 10$. $\lambda_{\text{lasso}} = 15$ with $\delta = \log 2 / (9 \lambda_{\text{lasso}}) = 0.005134$, so a stays in $[0.50, 1]$ at every λ (raising λ also sharpens the smoothing -- the two are coupled by that choice). Step $\eta = 1e-05$. $R = 100$ independent replicates on one fixed stratified 80/20 split ($n_{\text{train}} = 15216$, $n_{\text{test}} = 3804$). Bands: mean $\pm$ 1 SD (ddof = 1) -- repeat-run variability at a fixed split, not a confidence or credible interval. All λ are reported in results/lambda/lambda_lasso.csv.